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DIY Home NAS using Old PC

A Low-Cost Personal Cloud Solution

Problem & Motivation

- **Problem:**
  - Limited cloud storage (e.g., 15 GB Google Drive)
  - Multiple devices with scattered data
  - Need for secure and centralized storage
- **Motivation:**
  - Reuse old unused PC
  - Avoid subscription costs
  - Learn real-world system design

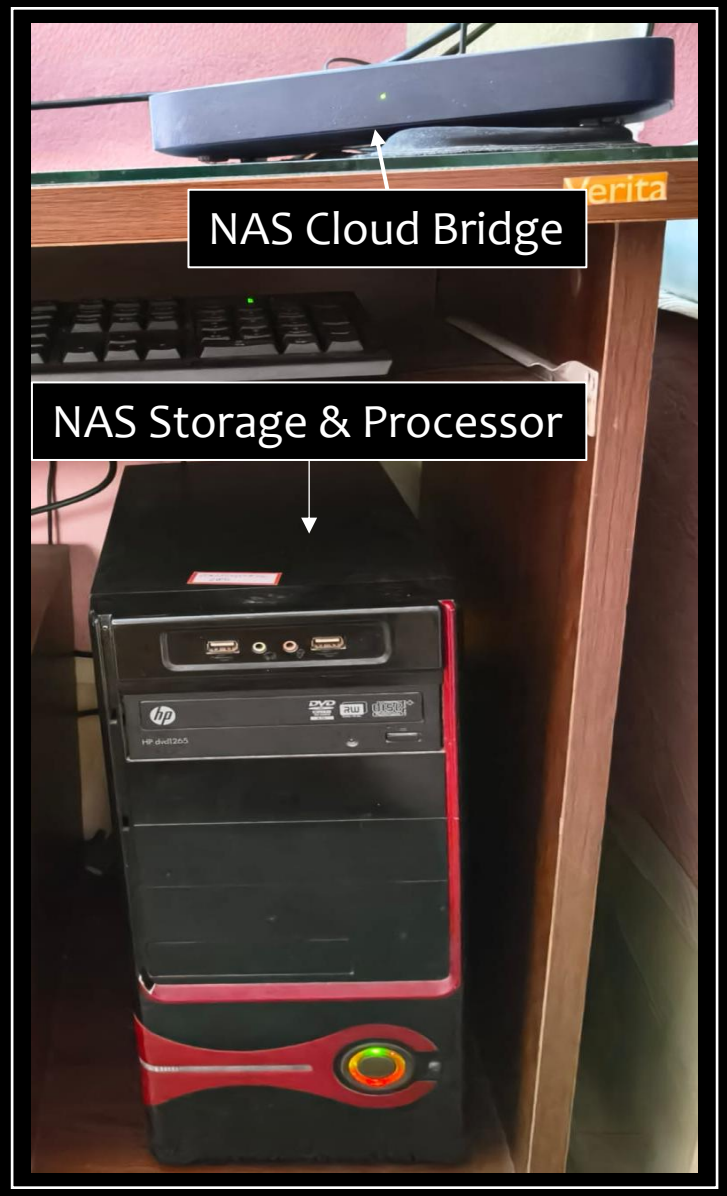
Key Idea:

Convert old hardware into a personal cloud server

Hardware Setup

System Configuration:

- Intel Pentium Processor (LGA1155)
- 12 GB DDR3 RAM
- 256 GB SSD (OS & Apps)
- 1 TB HDD (Storage)
- 450W PSU



Hardware Setup (in Working Condition)

Software Architecture

Operating Platform:

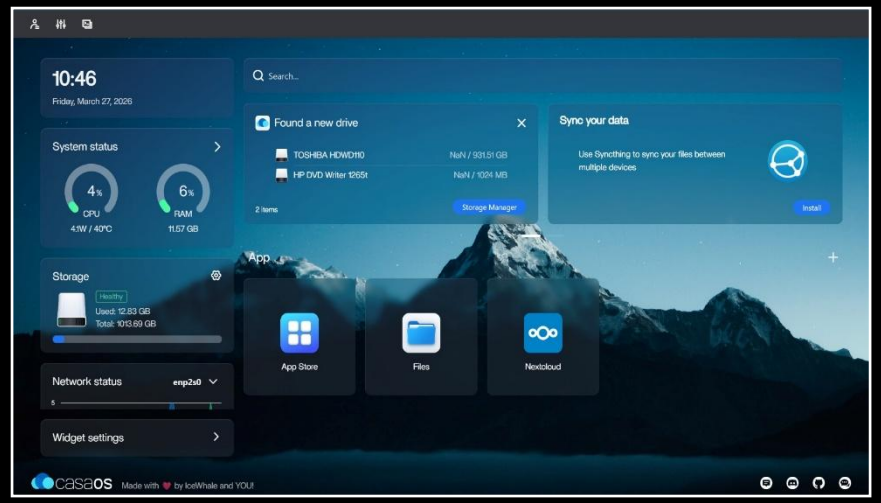
- CasaOS (Lightweight NAS OS)

Core Tools Used:

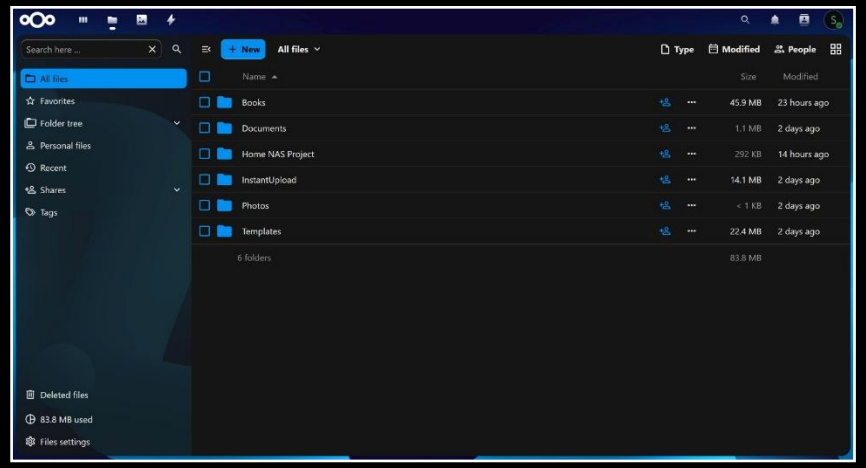
- Nextcloud → File storage & access
- Tailscale → Secure remote access

Features:

- Web-based interface
- App-based ecosystem
- Easy management



CasaOS Interface



NextCloud Interface

Features & Functionality

Key Features:

- Remote access from anywhere
- Automatic photo & video backup
- Manual document upload
- Multi-user support (family access)
- Secure VPN-based connection

Performance:

- Local access → High speed
- Remote access → Limited by 30 Mbps upload

User-Friendly:

Simple UI for non-technical users

Advantages & Learning Outcomes

Advantages:

- Cost-effective (uses existing hardware)
- Full data ownership
- No subscription required
- Scalable and customizable

Learning Outcomes:

- Networking (remote access, bandwidth limits)
- Linux & server management
- Storage systems & backup strategies
- Real-world system design

Conclusion: A practical and efficient personal cloud solution built from old hardware

Thank You